→ complex functions are eigenfunctions of LTI systems:

$$y(t) = \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau = \int_{-\infty}^{\infty} h(\tau)\, e^{st-s\tau}\, d\tau$$

$$= x(t) \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau = x(t)\, H(s)$$

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]\, x[n-k] = z^n \sum_{k=-\infty}^{\infty} h[k] z^{-k}$$

$$= x[n] \cdot H(z)$$

⇒ let $x(t) = a_1 e^{s_1 t} + a_2 e^{s_2 t} + a_3 e^{s_3 t}$

$$\Rightarrow y(t) = a_1 H(s_1) e^{s_1 t} + \dots$$

$$= \sum_k a_k H(s_k) e^{s_k t}$$

similarly: $y[n] = \sum_k a_k H(z_k) z_k^n$

$y(t) = x(t-3)$

$x(t) = \cos 4t + \cos 7t$

$\Rightarrow y(t) = \cos(4t-12) + \cos(7t-21)$

synthesis equation

$$\Rightarrow x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk\cdot 2\pi t/T}$$

→ fourier series representation

↓

periodic with period = T

$k=0$ term: const

$k=\pm 1$ term: freq $\omega_0 \Rightarrow$ fundamental components

$\Rightarrow$ if $x(t)$ is real: $a_k = a_{-k}^*$

$$\Rightarrow x(t) = a_0 + \sum_{k=1}^{\infty} \left[a_k e^{jk\omega_0 t} + a_k^* e^{-jk\omega_0 t} \right]$$

$$= a_0 + \sum_{k=1}^{\infty} 2\,\mathrm{Re}(a_k e^{jk\omega_0 t})$$

if $a_k = A_k e^{j\theta_k} \Rightarrow x(t) = a_0 + 2\sum_{k=1}^{\infty} A_k \cos(k\omega_0 t + \theta_k)$

if $a_k = B_k + jC_k \Rightarrow x(t) = a_0 + 2\sum_{k=1}^{\infty} [B_k \cos k\omega_0 t - C_k \sin k\omega_0 t]$

$$\Rightarrow x(t) e^{-jn\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} e^{-jn\omega_0 t}$$

$$\int_0^T x(t) e^{-jn\omega_0 t}\, dt = \sum_{k=-\infty}^{\infty} a_k \int_0^T e^{j\omega_0 t(k-n)}\, dt$$

T if $k=n$, 0 if $k \neq n$

$$1/T \int_T x(t) e^{-jn\omega_0 t}\, dt = a_n$$

→ analysis eq^n fourier series coefficients

$$x(t) = \sum_k a_k e^{jk\omega_0 t}$$

$$x(t) = 1 + \sin\omega_0 t + 2\cos\omega_0 t + \cos\left(2\omega_0 t + \frac{\pi}{4}\right)$$

$$= 1 + \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j} + 2\,\frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} + \frac{e^{j(2\omega_0 t + \pi/4)} + e^{-j(2\omega_0 t + \pi/4)}}{2}$$

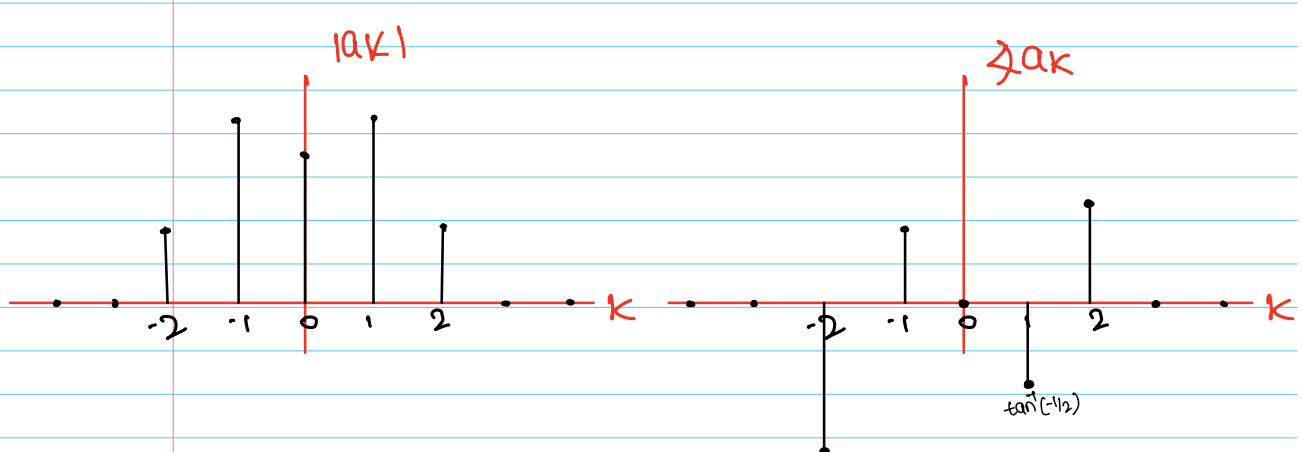
$$= 1 + e^{j\omega_0 t}\left[1 + \frac{1}{2j}\right] + e^{-j\omega_0 t}\left[1 - \frac{1}{2j}\right] + e^{j\cdot 2\omega_0 t}\left[\frac{e^{j\pi/4}}{2}\right] + e^{-j\cdot 2\omega_0 t}\left[\frac{e^{j\pi/4}}{2}\right]$$

$a_0 = 1$

$a_1 = 1 + 1/2j = 1 - j/2 \qquad a_{-1} = 1 - \frac{1}{2j} = 1 + j/2$

$a_2 = e^{j\pi/4}/2 = 1/2\sqrt{2}\,(1+j) \qquad a_{-2} = 1/2\sqrt{2}\,[1-j]$

other $a_k = 0$



square wave

$$x(t) = \begin{cases} 1 & |t| < T_1 \\ 0 & T_1 < |t| < T/2 \end{cases} \qquad \text{period} = T$$

$$a_0 = \frac{1}{T}\int_{-T_1}^{T_1} dt = \frac{2T_1}{T}$$

$$a_n = \frac{1}{T}\int_{-T_1}^{T_1} e^{-jn\omega_0 t}\, dt = \frac{-1}{jn\omega_0 T}\left(e^{-jn\omega_0 T_1} - e^{jn\omega_0 T_1}\right)$$

$$= \frac{2j\sin(\omega_0 n T_1)}{jn\omega_0 T} = \frac{\sin(\omega_0 n T_1)}{n\pi}$$

50% duty cycle $\Rightarrow T = 4T_1 \Rightarrow a_k = \frac{\sin(\pi k/2)}{\pi k}$

$\Rightarrow$ conditions for signals to be represented by fourier:

$\int_T |x(t)|^2 dt < \infty$ (guarentees a_k is finite)

→ absolutely integrable

$\int_T |x(t)| dt < \infty$ (dirichlet cond')

→ In any finite time interval, $x(t)$ is of bounded variation i.e no more than finite maxima & minima during

any single period of the signal

→ only finite number of discontinuities, each of these discontinuities is finite

⇒ Properties of continuous time FS

→ Linearity:

$x(t) \to a_k$ } both x and y
$y(t) \to b_k$ } have time period T

$z(t) = Ax(t) + By(t) \longrightarrow c_k = Aa_k + Bb_k$

→ Time shifting:

$x(t) \to a_k$

$x(t-t_0)$:

$$b_k = \frac{1}{T}\int_T x(t-t_0)\, e^{-jk\omega_0 t}\, dt$$

$\tau = t - t_0 \Rightarrow b_k = \frac{1}{T}\left(\int_T x(t)\, e^{-jk\omega_0 \tau}\, d\tau\right)\left(e^{-jk\omega_0 t_0}\right)$

$$b_k = a_k e^{-jk\omega_0 t_0}$$

$|b_k| = |a_k|$

→ Time reversal: $x(t) \rightarrow a_k$

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$$x(-t) = \sum_{k=-\infty}^{\infty} a_k e^{-jk\omega_0 t} = \sum_{l=-\infty}^{\infty} a_{-l} e^{jl\omega_0 t} \qquad k = -l$$

$$\Rightarrow x(-t) \longrightarrow a_{-k}$$

$x(t) \rightarrow$ even $\Rightarrow a_{-k} = a_k$

$x(t) \rightarrow$ odd $\Rightarrow a_{-k} = -a_k$

→ Time scaling:

$$x(\alpha t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 \alpha t}$$

fourier series itself has changed

→ Multiplication:

$x(t) \rightarrow a_k \qquad y(t) \rightarrow b_k$

$x(t) \cdot y(t) \rightarrow \sum_{l=-\infty}^{\infty} a_l b_{k-l}$ → discrete time convolution

→ Conjugation:

$x(t) \Rightarrow a_k \Rightarrow x^*(t) \rightarrow a_{-k}^*$

$x(t) \rightarrow$ real $\Rightarrow \quad a_{-k}^* = a_k$

→ Parseval theorm:

$$\frac{1}{T}\int_T |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |a_k|^2$$

average power in 1 time period

→ Frequency shift:

$$x(t) = \sum a_k e^{jk\omega_0 t}$$

$$y(t) = x(t)\cdot e^{jm\omega_0 t} \rightarrow a_{k-m}$$

→ Periodic convolution:

$$x(t) \rightarrow a_k \qquad y(t) \rightarrow b_k$$

$$\int_T x(\tau)y(t-\tau)\,d\tau \rightarrow T a_k b_k$$

→ Differentiation: $jk\omega_0 a_k$

→ integration: $a_k/jk\omega_0 \leftarrow \int_{-\infty}^{t} x(t)dt$

finite valued & periodic only if $a_0 = 0$

Suppose we are given the following facts about a signal $x(t)$:

1. $x(t)$ is a real signal.
2. $x(t)$ is periodic with period $T = 4$, and it has Fourier series coefficients a_k.
3. $a_k = 0$ for $|k| > 1$.
4. The signal with Fourier coefficients $b_k = e^{-j\pi k/2}a_{-k}$ is odd.
5. $\frac{1}{4}\int_4 |x(t)|^2 dt = 1/2$.

$\omega_0 = \pi/2$

condition 3 $\Rightarrow a_0, a_1, a_{-1}$ only

$\Rightarrow x(t) = a_0 + a_1 e^{j\omega_0 t} + a_{-1} e^{-j\omega_0 t}$

$|a_0|^2 + |a_1|^2 + |a_2|^2 = \frac{1}{2}$

$x^*(t) = a_0^* + a_1^* e^{-j\omega_0 t} + a_{-1}^* e^{j\omega_0 t}$

$x(t) = x^*(t) \Rightarrow a_0 = a_0^*$

$a_1^* = a_{-1}$ $\quad a_{-1}^* = a_1$

$\Rightarrow a_0, a_1, a_{-1}$ real

$\Rightarrow a_1 = a_{-1}$

$b_k = e^{-j\pi k/2} a_{-k}$

$y(t) = \sum e^{-j\pi k/2} a_{-k} e^{jk\omega_0 t}$

$= \sum a_{-k} e^{j\pi k/2 (t-1)}$

$y(t) = a_0 + a_{-1} e^{j\pi/2 (t-1)} + a_1 e^{-j\pi/2 (t-1)}$

$y(-t) = a_0 + a_{-1} e^{-j\pi/2 (t+1)} + a_1 e^{j\pi/2 (t+1)}$

$y(t) = -y(-t) \Rightarrow a_0 = -a_0 \Rightarrow a_0 = 0$

$-a_{-1} e^{-j\pi/2} = a_1 e^{j\pi/2}$

$\Rightarrow -a_{-1}(-j) = a_1 (j) \Rightarrow a_1 = a_{-1}$

$\Rightarrow a_1 = 1/2 \quad a_{-1} = 1/2$ or $a_1 = -1/2 \quad a_{-1} = 1/2$

$\Rightarrow x(t) = \pm \cos(\pi t/2)$

$\Rightarrow$ Discrete time signal:

$$x[n] = x[n+N]$$

$$\phi_k[n] = e^{jk\omega_0 n} = e^{jk(2\pi/N)n} \qquad k = 0, \pm 1, \ldots$$

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}$$

→ discrete time fourier series

$\langle N \rangle = 0$ to $N-1$ or 3 to $N+2$ or ....

$$x[0] = \sum_{k=\langle N \rangle} a_k \qquad x[N-1] = \sum_{k=\langle N \rangle} a_k e^{jk2\pi(N-1)/N}$$

$$x[1] = \sum_{k=\langle N \rangle} a_k e^{j2\pi k/N}$$

$$\sum_{n=\langle N \rangle} x[n] e^{-jr2\pi n/N} = \sum_{k=\langle N \rangle} a_k \sum_{n=\langle N \rangle} e^{j(k-r)\cdot 2\pi n/N}$$

$k = r \Rightarrow 1/N$ (else 0)

$$\Rightarrow a_r = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jr(2\pi n/N)}$$

$$\boxed{a_r = a_{r+N}}$$

$N \rightarrow$ integer

$x[n] = \sin \omega_0 n \qquad \omega_0 = 2\pi/N$

$$x[n] = \frac{1}{2j} e^{j\,2\pi/N\, n} - \frac{1}{2j} e^{-j\cdot 2\pi/N\, n}$$

$a_1 = 1/2j$ ($= a_{N+1}$) $\quad a_{-1} = -1/2j$ ($= a_{N-1}$)

N → rational

$$\Rightarrow x[n] = \frac{1}{2j} e^{j 2\pi (M/N) n} - \frac{1}{2j} e^{-2\pi (M/N) n}$$

$$a_M = a_{M+N} = 1/2j \qquad a_{-M} = a_{M-N} = \frac{-1}{2j}$$

In this example, we consider the discrete-time periodic square wave. We can evaluate the Fourier series for this signal. Because x[n] = 1 for - N1 < n < N1, it is particularly convenient to choose the length-N interval of summation

$$a_k = \frac{1}{N} \sum_{n=-N_1}^{N_1} e^{-jk 2\pi/N n} \qquad k = 0, \pm N_1, \pm 2N \cdots$$

$$a_k = \frac{2N_1+1}{N} \rightarrow a_k = \frac{1}{N} \sum_{-N_1}^{N_1} \left(e^{-j\frac{2\pi k}{N}} \right)^n$$

$$= \frac{1}{N} \frac{e^{+j\left(\frac{2\pi k N_1}{N}\right)} \left[\left(e^{-j \cdot 2\pi k/N (2N_1+1)} - 1 \right) \right]}{e^{-j \cdot 2\pi k/N} - 1}$$

$$= \frac{1}{N} \frac{\left[e^{-j \cdot 2\pi k/N (N_1+1)} - e^{j \cdot 2\pi k/N (N_1)} \right]}{e^{-j \cdot 2\pi k/N} - 1}$$

$$= \frac{1}{N} \frac{\sin\left[2\pi k (N_1 + 1/2)/N \right]}{\sin(\pi k/N)} \qquad k \neq 0, N, 2N \ldots$$

⇒ Properties:

TABLE 3.2 PROPERTIES OF DISCRETE-TIME FOURIER SERIES

Property	Periodic Signal	Fourier Series Coefficients
	$x[n]$, $y[n]$ } Periodic with period N and fundamental frequency $\omega_0 = 2\pi/N$	a_k, b_k } Periodic with period N
Linearity	$Ax[n] + By[n]$	$Aa_k + Bb_k$
Time Shifting	$x[n - n_0]$	$a_k e^{-jk(2\pi/N)n_0}$
Frequency Shifting	$e^{jM(2\pi/N)n}x[n]$	a_{k-M}
Conjugation	$x^*[n]$	a^*_{-k}
Time Reversal	$x[-n]$	a_{-k}
Time Scaling	$x_{(m)}[n] = \begin{cases} x[n/m], & \text{if } n \text{ is a multiple of } m \\ 0, & \text{if } n \text{ is not a multiple of } m \end{cases}$ (periodic with period mN)	$\frac{1}{m}a_k$ (viewed as periodic with period mN)
Periodic Convolution	$\sum_{r=\langle N \rangle} x[r]y[n-r]$	Na_kb_k
Multiplication	$x[n]y[n]$	$\sum_{l=\langle N \rangle} a_l b_{k-l}$
First Difference → diff	$x[n] - x[n-1]$	$(1 - e^{-jk(2\pi/N)})a_k$
Running Sum → integral	$\sum_{k=-\infty}^{n} x[k]$ (finite valued and periodic only if $a_0 = 0$)	$\left(\frac{1}{(1 - e^{-jk(2\pi/N)})}\right)a_k$
Conjugate Symmetry for Real Signals	$x[n]$ real	$\begin{cases} a_k = a^*_{-k} \\ \Re e\{a_k\} = \Re e\{a_{-k}\} \\ \Im m\{a_k\} = -\Im m\{a_{-k}\} \\ \lvert a_k\rvert = \lvert a_{-k}\rvert \\ \measuredangle a_k = -\measuredangle a_{-k} \end{cases}$
Real and Even Signals	$x[n]$ real and even	a_k real and even
Real and Odd Signals	$x[n]$ real and odd	a_k purely imaginary and odd
Even-Odd Decomposition of Real Signals	$\begin{cases} x_e[n] = \mathcal{E}v\{x[n]\} & [x[n] \text{ real}] \\ x_o[n] = \mathcal{O}d\{x[n]\} & [x[n] \text{ real}] \end{cases}$	$\begin{cases} \Re e\{a_k\} \\ j\Im m\{a_k\} \end{cases}$

Parseval's Relation for Periodic Signals

$$\frac{1}{N}\sum_{n=\langle N \rangle} |x[n]|^2 = \sum_{k=\langle N \rangle} |a_k|^2$$

→ Fourier series and LTI systems:

If $x(t) = e^{st} \Rightarrow y(t) = H(s)e^{st}$

$$H(s) = \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau$$

$x[n] = z^n \Rightarrow y[n] = H(z) z^n$

$$H_z = \sum_{k=-\infty}^{\infty} h[k] z^{-k}$$

This course we focus on $s = j\omega$ ($\mathrm{Re}(s) = 0$) i.e input is a complex sinusoid at frequency ω)

$$H(j\omega) = \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau$$

↳ frequency response of the system

Similarly, for discrete, we focus on $z = e^{j\omega}$ and $H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h[k] e^{-j\omega k}$

$$\rightarrow x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$$\Rightarrow y(t) = \sum_{k=-\infty}^{\infty} a_k H(jk\omega_0) e^{jk\omega_0 t}$$

$h(t) = e^{-t} u(t)$

$$\Rightarrow H(j\omega) = \int_{-\infty}^{\infty} e^{-\tau} u(t) e^{-j\omega\tau} d\tau$$

$$= 1/(1+j\omega)$$

$$h[n] = \alpha^n u[n]$$

$$x[n] = \cos\left(\frac{2\pi n}{N}\right)$$

$$\Rightarrow x[n] = \frac{1}{2} e^{j \cdot 2\pi n/N} + \frac{1}{2} e^{-j \cdot 2\pi n/N}$$

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} \alpha^k u[k] e^{-j\omega k}$$

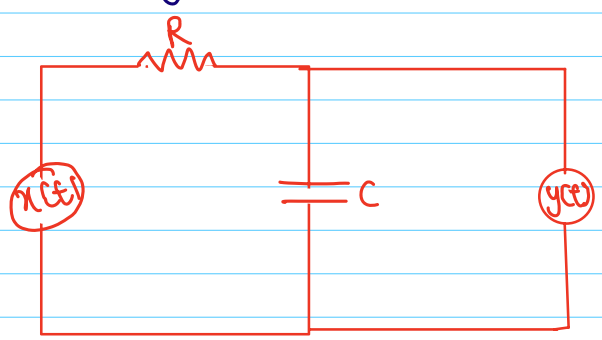
$$= \sum_{k=0}^{\infty} (\alpha e^{j\omega})^k = \frac{1}{1 - \alpha e^{-j\omega}}$$

$$y[n] = \sum_{k=-\infty}^{\infty} a_k H(e^{jk\omega_0}) \cdot e^{jk\omega_0 n}$$

$$= \frac{1}{2} H(e^{j \cdot 2\pi/N}) e^{j 2\pi/N n} + 1/2 H(e^{-j \cdot 2\pi/N}) e^{-j \cdot 2\pi/N n}$$

$$= \frac{1}{2} \frac{1}{1 - \alpha e^{j 2\pi/N}} e^{j \cdot \frac{2\pi n}{N}} + \frac{1}{2} \frac{1}{1 - \alpha e^{2\pi j/N}} e^{-j \cdot \frac{2\pi n}{N}}$$

⇒ Filtering: changing relative amplitudes of frequency components in a signal



$iR + \frac{q}{C} = x(t)$ $\qquad$ $y(t) = q$

$$y'(t)RC + y(t) = x(t)$$

$$RC\, j\omega e^{j\omega t} H(j\omega) + e^{j\omega t} H(j\omega) = e^{j\omega t}$$

$$H(j\omega) = \frac{1}{1 + j\omega RC}$$